**Unit:** 6 Unit 8: Quadratics
**Topic:** Definition of Quadratics and Standard Form

**Name:** ________________
**Date:** ____________

# Advanced (Habanero)

**Q1.** What is the standard form of a quadratic function?
(A) $ax^2 + bx + c$
(B) $ax + bx^2 + c$
(C) $ax^2 - bx + c$
(D) $ax^3 + bx^2 + c$
(E) $ax^2 + b$

**Q2.** Which of the following is a quadratic function in standard form?
(A) $x^2 + 3x - 2$
(B) $x^3 - 2x^2 + x$
(C) $x^2 - 4$
(D) $x^4 + 2x^2 - 1$
(E) $x - 2$

**Q3.** What is the value of $a$ in the quadratic function $2x^2+3x-1$?
(A) 1
(B) 2
(C) 3
(D) $-1$
(E) $-2$

**Q4.** Which of the following quadratic functions has a vertex at $(0, 0)$?
(A) $x^2 + 2$
(B) $x^2 - 2$
(C) $x^2$
(D) $-x^2$
(E) $x^2 + 1$

**Q5.** What is the axis of symmetry of the quadratic function $x^2 + 2x + 1$?
(A) $x = -1$
(B) $x = 1$
(C) $x = 0$
(D) $x = -2$
(E) $x = 2$

**Q6.** Which of the following is a key feature of the quadratic function $x^2 + 2x + 1$?
(A) Vertex at $(0, 0)$
(B) Axis of symmetry at $x = 0$
(C) Vertex at $(-1, 0)$
(D) Axis of symmetry at $x = -1$
(E) Minimum value at $x = 0$

**Q7.** What is the minimum value of the quadratic function $x^2 + 2x + 1$?
(A) 0
(B) 1
(C) $-1$
(D) 2
(E) $-2$

**Q8.** Which of the following quadratic functions has a maximum value?
(A) $x^2 + 2$
(B) $-x^2 + 2$
(C) $x^2 - 2$
(D) $-x^2 - 2$
(E) $x^2$

## Open Ended Questions (FRQ)

**Q9.** Find the vertex and axis of symmetry of the quadratic function $2x^2 + 4x + 1$. Show all work and explain your reasoning.

**Q10.** Graph the quadratic function $x^2 - 2x - 1$ and identify its key features, including vertex, axis of symmetry, and minimum value. Explain how you determined each feature.

**Chef's Tips**



- Emphasize the importance of identifying key features of quadratic functions
- Use graphing tools to visualize quadratic functions and their key features
- Encourage students to show all work and explain their reasoning